UNIVERSITAT DE BARCELONA

ICCUB | Institut de Ciències del Cosmos UNIVERSITAT DE BARCELONA

EXCELENCIA MARÍA DE MAEZTU 2020-2023

Master in Astrophysics, Particle Physics, and Cosmology

Academic year 2024-2025

Star Formation and Structure

Fall semester: Mon, Tue, Wed, Thur, 17:40h – 18:40h
Room V12M

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Rosario López (rosario@fqa.ub.edu)
Nadejda Blagorodnova (nblago@fqa.ub.edu)









Lines of Research in the Star Formation Group

- High-angular resolution observations of the first stages of stellar evolution
- Outflows, jets, and accretion disks in low- and high-mass young stellar objects
- Computational models of star-forming clouds and star formation
- Interstellar-medium turbulence

Institute of Space Sciences

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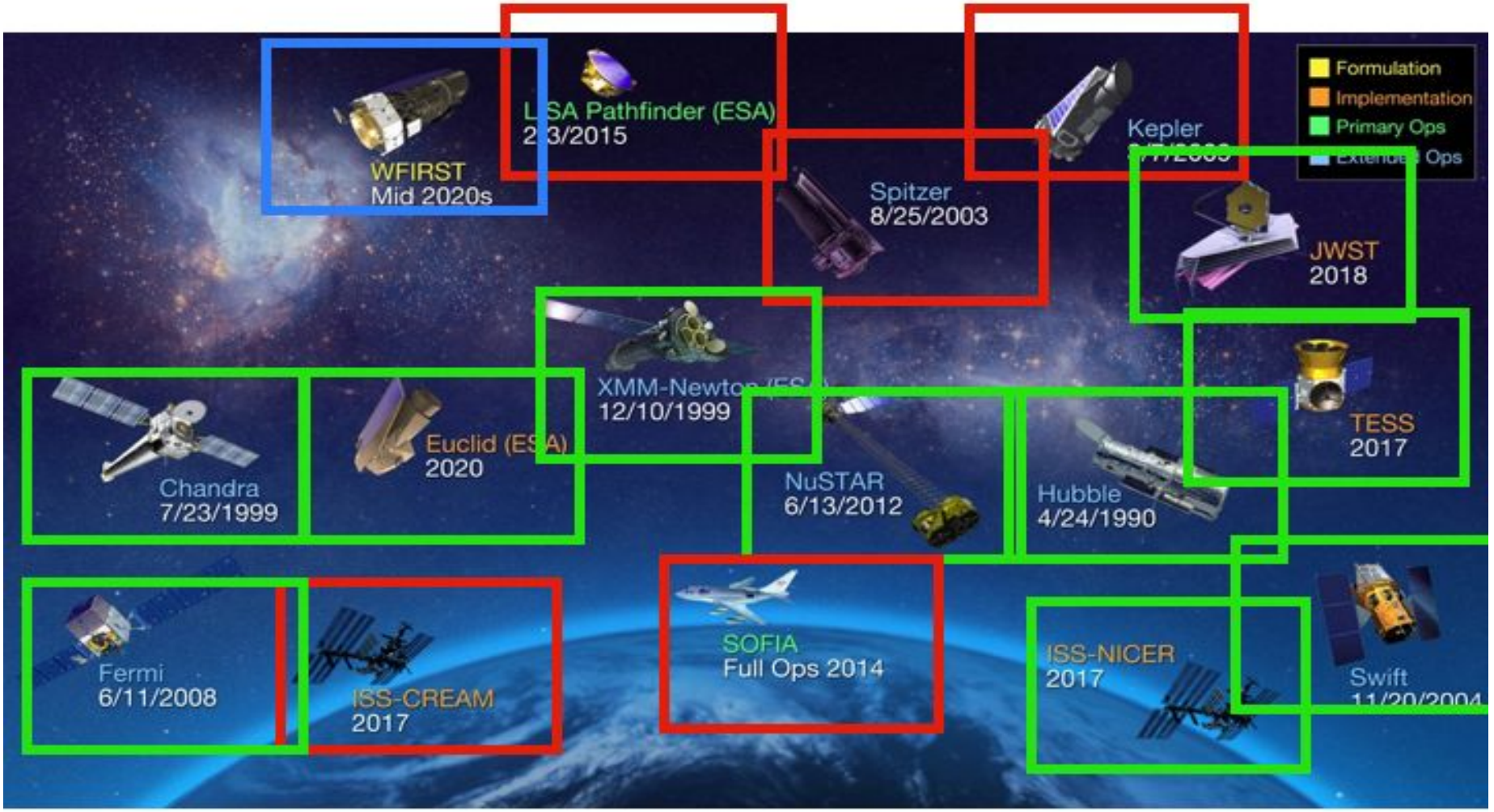

Lines of Research in the Stellar Evolution Group

- Observational study of different stages of binary stellar evolution:
  - Interacting binary systems and variability studies in large time-domain surveys.
  - Astrophysical transients resulting from binary interaction: stellar mergers and interacting supernovae.
  - Binary interaction remnants: geometries and dust formation
  - Connection of binary remnants to GW sources

MULTI-WAVELENGTH OBSERVATIONS





How do stars form and evolve?

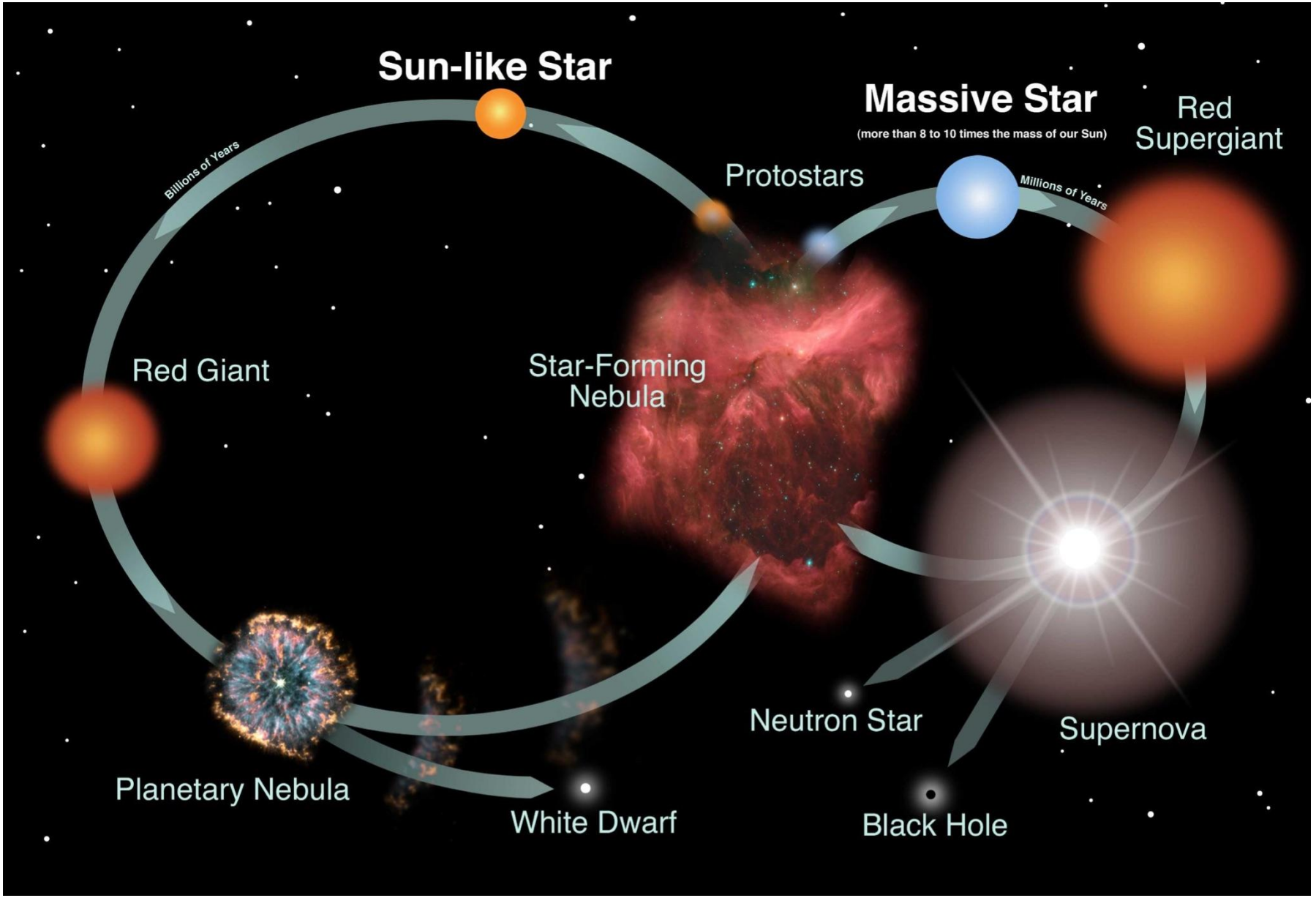
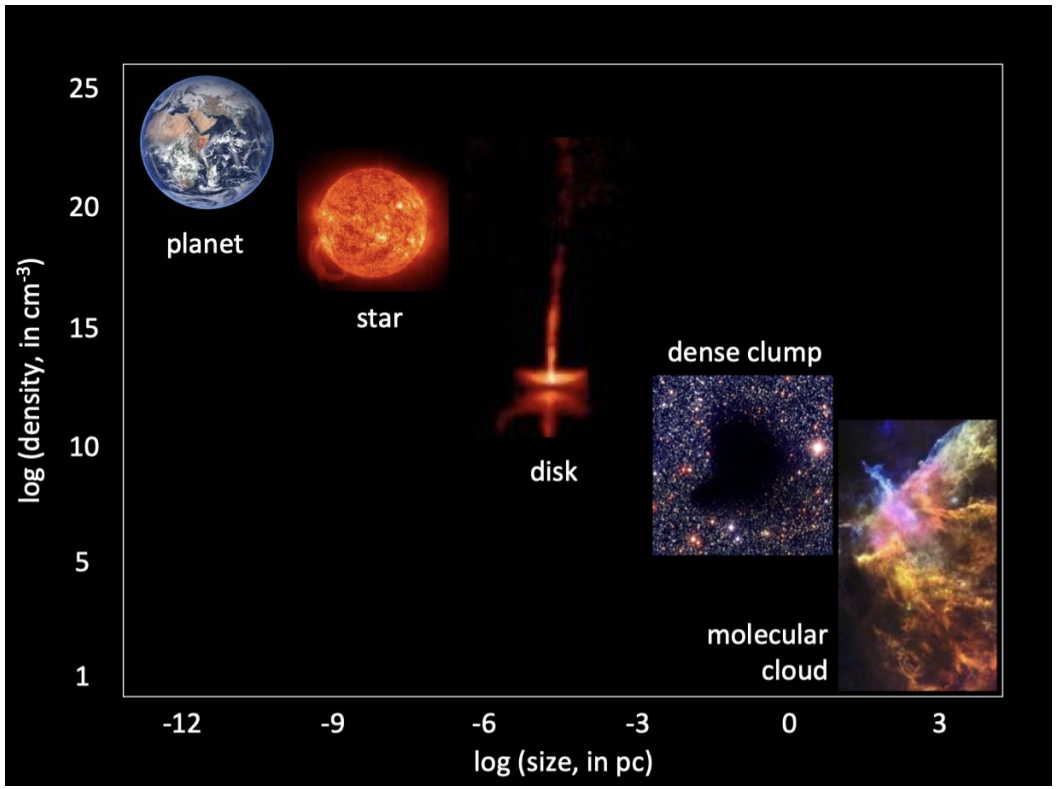


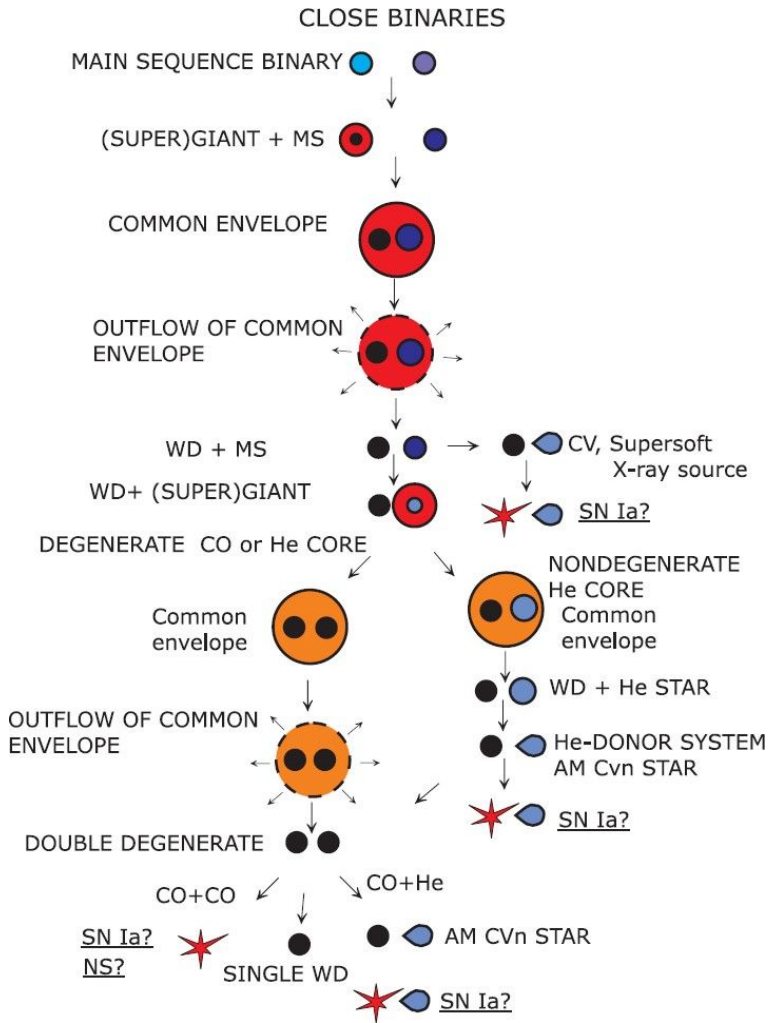
Image Credit: *NASA and the Night Sky Network.*

OBJECTIVES

To understand the physics of the ISM, the star formation process in our Galaxy and the pre-main sequence evolution of stars with different masses



To understand the evolutionary channels of binary stars, and their final fate.



TEACHING BLOCKS

1. Introduction Gemma

- The Interstellar Medium: dust and gas

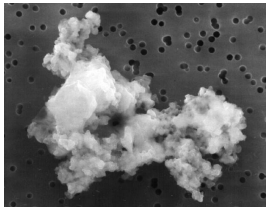
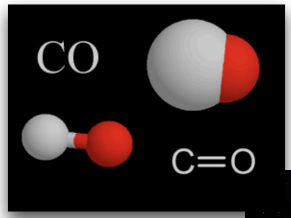


Image Credit: NASA

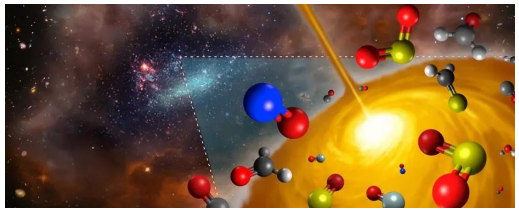




Credit: R. Friesen, Dunlap Institute/J. Pineda, MPIP /GBO /AUI /NSF /NASA

2. Star-forming regions Gemma

- Molecular clouds: energy balance, cloud collapse and fragmentation
- The Initial Mass Function
- Low-mass versus high-mass star formation
- Astrochemistry



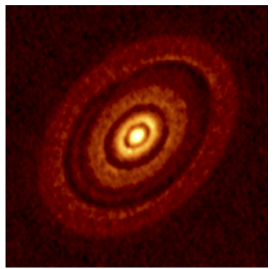
Credit: ALMA Observatory

3. Young Stellar Objects Rosario

- Spectral Energy Distribution
- Pre-main sequence evolution
- T Tauri stars and Ae/Be stars
- Interaction of YSOs with their environment: jets, HH objects and molecular outflows
- Accretion and supersonic ejection processes
- Protoplanetary disks Gemma



Image Credit: ESA/Webb, NASA, CSA, Tom Ray (Dublin)



Credit: ALMA (ESO/NAOJ/ NRAO)

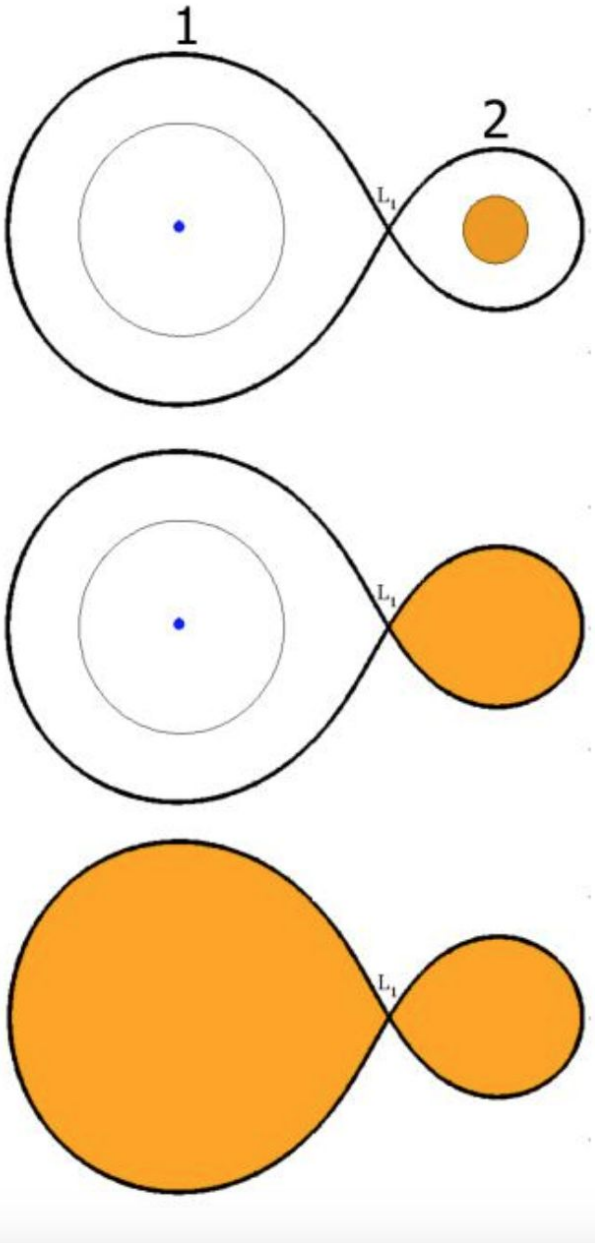
TEACHING BLOCKS

4. Stellar Evolution Nadia

- Stellar structure
- Energy generation and transport
- Stellar timescales
- Single stellar evolution: low-mass vs-. high-mass

5. Binary systems Nadia

- Roche Lobe geometry
- Types of binaries
- Mass transfer and common envelope evolution
- Binary remnants and GW sources



TEACHING BLOCKS

6. The Tools

- Radio astronomy Gemma
- Optical and near-infrared astronomy Rosario
- MESA stellar evolution code Nadia





Practical cases

Gemma

Proposal writing

Basic concepts on AOT, calibration and imaging using CASA

Nadia

Evolution of a Sun-like star in MESA



ALMA Observing Tool

CASA
Common Astronomy Software Applications

MESA

INVITED LECTURES

- **Álvaro Sánchez-Monge**

Formation of high-mass stars





Credit: NASA/ESA

Institute of Space Sciences

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EVALUATION

6 ECTS: 50 hours (lectures, hands-on activities and discussions)

Work required to the students

- Class attendance and active participation
- Small exercises to be presented during the course
- Hands-on activities: analysis of ALMA data, running stellar evolution models
- Elaboration of observational proposal
- Discussion of scientific papers



https://campusvirtual.ub.edu

50% of the final mark

- Written examination: short-answer questions

50% of the final mark

Master Thesis Proposals

(https://icc.ub.edu/master_afpc/thesis/astro)

Spectroscopic characterization of stellar merger candidate systems

Advisor: Nadia Blagorodnova



OTHER IDEAS

Developing line identification algorithms in the era of Large Surveys

Advisor: Gemma Busquet

Grain growth and chemical composition in G14.225-0.506

Advisor: Gemma Busquet

Exploring Yellow Supergiants with large time-domain surveys

Advisor: Nadia Blagorodnova

Non-thermal emitters in Orion A

Advisors: Valentí Bosch-Ramon & Gemma Busquet

In collaboration: **High Energy Astrophysics Group**